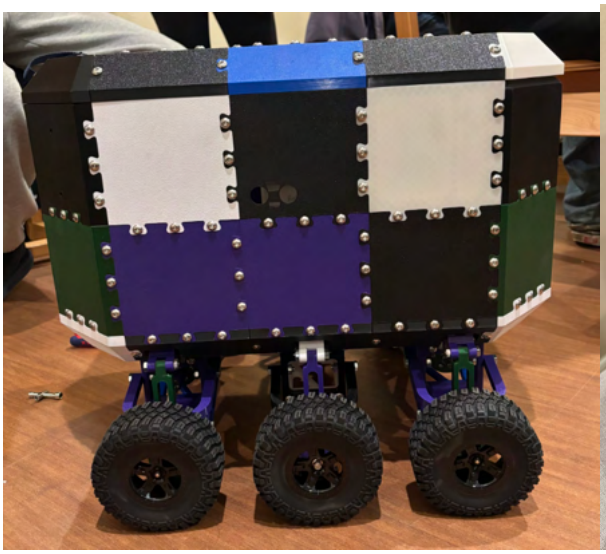
Mechanical Engineering Portfolio

Taehyun “Ted” Um

Mechanical Design | Electromechanical Integration | Rapid Prototyping | Hardware Validation







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FEATURED PROJECTS

Project	Role	Focus
Six-Axis Robotic Arm	Mechanical Design Lead	Gearboxes, packaging, materials, prototyping
Semi-Autonomous Delivery Robot	Design Team Lead	Modular chassis, integration, validation
Full-Scale Go-Kart	Designer and Fabricator	Chassis, welding, drivetrain

Portfolio Positioning

I am a hands-on mechanical engineering student who develops systems from requirements and CAD through fabrication, integration, and physical testing. My projects emphasize mechanical architecture, electromechanical packaging, additive manufacturing, design iteration, and root-cause troubleshooting.

Project Evidence Checklist

- Use original, high-resolution images and crop out distracting backgrounds.
- Label CAD views so a reviewer can understand the subsystem without reading the body text.
- Include at least one physical prototype, one test setup, and one meaningful iteration per project.
- State whether every number is targeted, calculated, simulated, or physically measured.
- Distinguish your individual contribution from the team’s result.
- Link to short demonstration videos when motion or control behavior matters.

Before Exporting

Last updated: 10/10/2026

Preferred role focus: ***Mechanical design, robotics hardware, manufacturing, prototype testing***

PROJECT 01

Six-Axis Robotic Arm

Mechanical Design Lead | Collaborative Personal Project | June 2026 to Present



Project Objective

Design and build a custom six-axis robotic arm using FDM-manufactured structures and integrated electromechanical joints. The current design targets approximately 650 mm of reach and a 400 g toolhead capacity while prioritizing serviceability, packaging efficiency, and repeatable manufacturing.

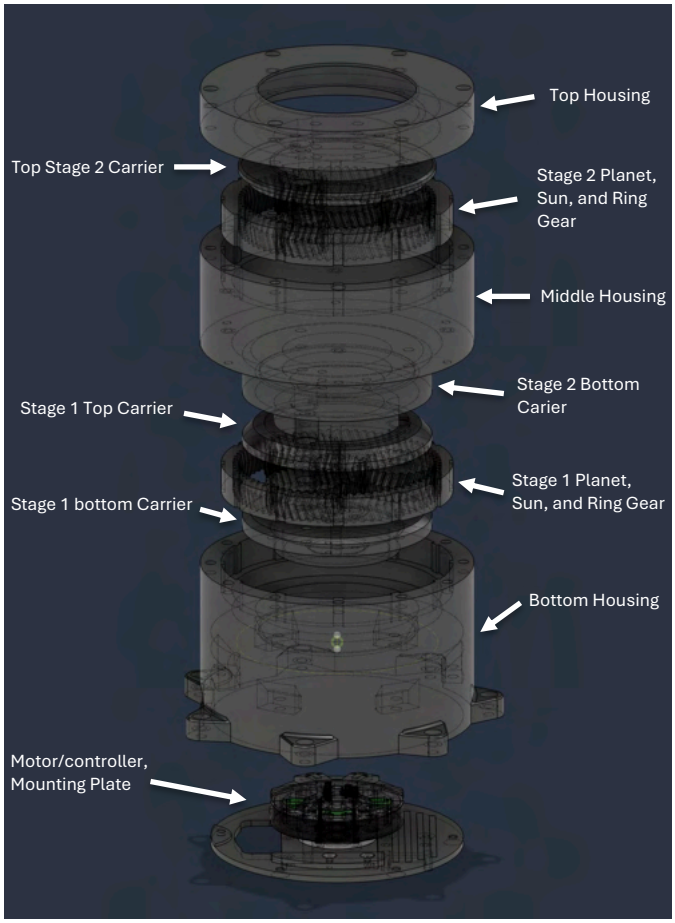
Requirement	Current Design Basis	Status
Reach	650 mm	Target
Toolhead capacity	400 g	Target
Joint 1 reduction	15:1 two-stage planetary	Designed and prototyped
Joint 2 reduction	20:1 cycloidal	Prototype development
Joint 1 output	18.4 N·m peak, 5.23 N·m continuous, approximately 431 rpm unloaded	Calculated
Control and sensing	Moteus r4.11 with input/output encoder integration, Nema Stepper motors with input/output encoders	Hardware integration

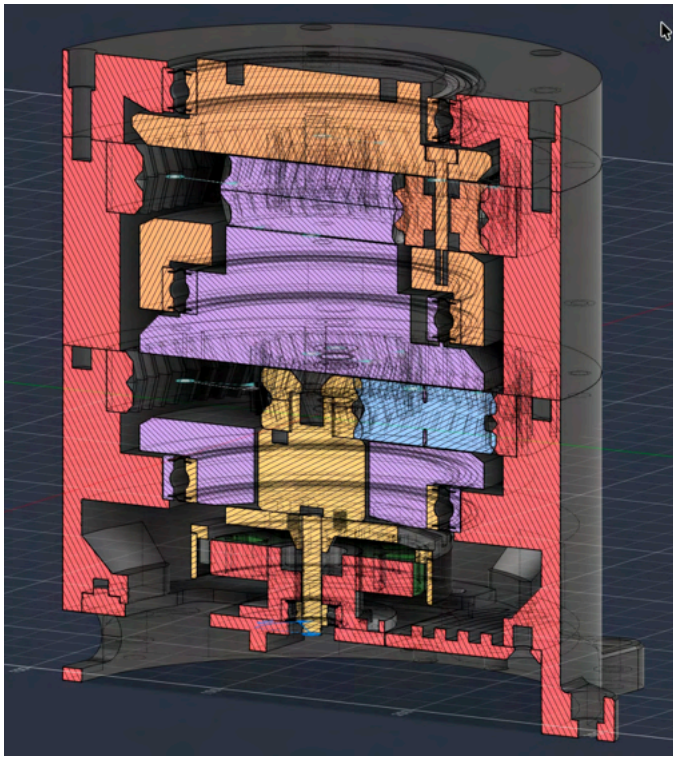
Current project status: Joint 1 completed, joint 2 in-progress

Primary design challenge: Developing a 15:1 herringbone planetary gearbox that could be assembled despite the opposing tooth angles. This required a split sun gear design with precise alignment between its two halves.

Project link: https://github.com/ssougu/6-DOF-ARM-PROJECT

Joint 1 Architecture and Design Decisions





Mechanical Architecture

Joint 1 packages an MJ5208 BLDC motor, Moteus r4.11 controller, magnetic encoder, bearings, and a 15:1 two-stage gearbox inside a serviceable printed housing. The gearbox combines a 5:1 first stage with a 3:1 second stage and uses three planets per stage.

Problem	Design Response	Why It Matters
Opposing herringbone angles prevent conventional one-piece assembly	Split the sun gear while maintaining tooth-phase alignment and resisting relative rotation	Preserves smooth tooth engagement and opposing axial thrust without sacrificing assembly
A damaged printed ring gear could require rebuilding the full housing	Made the nylon ring gear a separately replaceable insert	Reduces replacement printing from roughly nine hours to one
Tight joint volume must contain the drivetrain and electronics	Modeled around measured hardware dimensions and component interfaces	Improves fit, service access, and packaging confidence before printing

Trade study: I initially built a single-stage 5:1 spur planetary gearbox as a proof of concept because its simpler geometry made printing, assembly, and initial testing easier. I later developed a two-stage 15:1 herringbone gearbox to increase output torque while targeting smoother engagement, improved tooth load sharing, and reduced vibration. The opposing helix angles also balance axial thrust, although the geometry introduced additional printing and assembly challenges. I addressed the assembly constraint with a split sun gear whose halves install separately and lock together to maintain tooth alignment.

I originally considered a camera gimbal motor, but research and preliminary calculations indicated insufficient output for the intended application. I selected the MJ5208 because its documented performance and use in robotic applications

supported further design calculations and integration. Using estimated arm inertia, target motion, and drivetrain losses, I selected a 15:1 reduction to balance torque and speed while retaining additional design margin. Combining 5:1 and 3:1 stages achieved this reduction without the large diameter required by a single-stage 15:1 planetary gearbox.

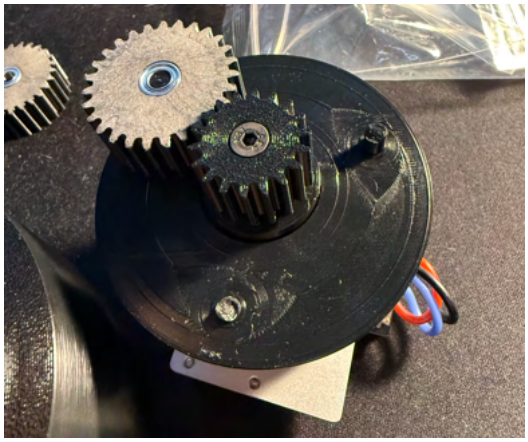
Tolerance strategy: I retained nominal hardware dimensions in CAD and applied 0.09 mm X-Y hole compensation in the slicer to adjust printed hole sizes. Throughout CAD development, I printed and tested individual components to verify MR105ZZ bearing fitment, hardware interfaces, and overall geometric accuracy before progressing to the complete assembly. Critical interfaces included bearing pockets, shoulder-bolt holes, split sun gear alignment, and the removable ring gear fit. These component-level checks allowed me to identify fitment issues early and refine the design through successive prototypes.

SIX-AXIS ROBOTIC ARM

Prototype Iteration and Manufacturing



More than 30 prototype prints and eight Joint 1 revisions have been used to investigate shrinkage, warping, drivetrain friction, preload, material behavior, support strategy, and component packaging. Testing has included PETG, nylon, PAHT-CF, and PET-CF components, with a replaceable nylon ring gear retained as a mechanical fuse.



Printed plastic planet pins fractured under radial loads during assembly and operation. I replaced them with stainless steel shoulder bolts, using the smooth shoulders to support the MR105ZZ planet bearings.



The nylon ring gear is a separately replaceable insert. External anti-rotation tabs engage the housing, while the retaining collar holds the insert axially. Separating the ring gear from the housing reduced replacement printing from approximately nine hours to one hour.

Observed Issue	Evidence	Design or Process Change
Broken printed plastic planet pins	*Pins fractured during assembly and operation indicating insufficient strength under the planet gears' radial loads.*	*Replaced printed pins with stainless steel shoulder bolts to provide stronger supports for MR105ZZ bearings.*
Gear-interface rubbing and binding	*Manual rotation checks identified rubbing that increased drivetrain resistance.*	*Added chamfers at gear interfaces to reduce interfaces to reduce interference, then checked the revised assembly for rotation without binding.*

Undersized printed holes	*Example observed: a nominal 10 mm hole printed approximately 9.72 mm*	*Retained nominal CAD dimensions and applied 0.09mm X-Y hole compensation in slicer. Printed and checked individual components throughout CAD development to verify bearing fit, hardware fit, and geometry.*

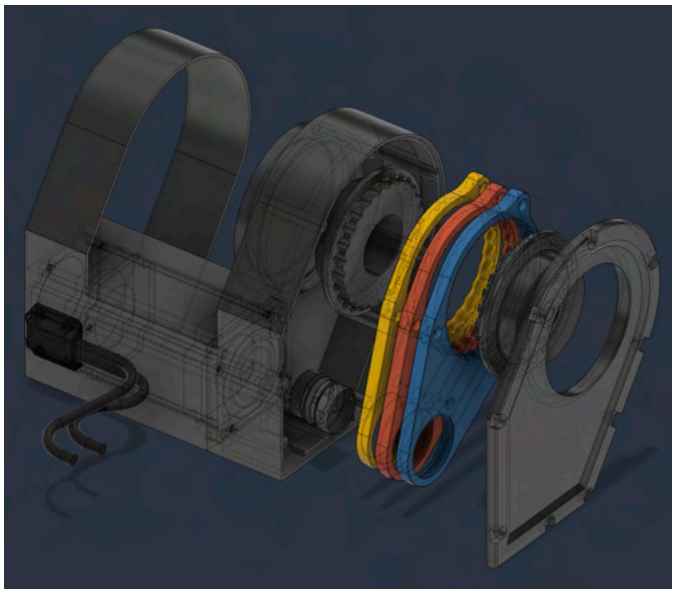
Final manufacturing configuration: I use Siraya Tech Fibreheart PET-CF for structural housings and brackets, and SUNLU 1.75 mm PA6/66 nylon for gears and mating components. Large housings use a 0.6 mm nozzle to shorten printing time, while smaller brackets and housings with precise hardware interfaces use a 0.4 mm nozzle. Nylon components are printed exclusively with a 0.4 mm nozzle.

Setting	PET-CF Small components	PET-CF Large housings	Nylon
Nozzle diameter	*0.4 mm*	*0.6 mm*	*0.4 mm*
Layer height	*0.12 mm*	*0.30 mm*	*0.20 mm*
Walls	*5*	*5*	*5*
Infill	*25% gyroid*	*25% gyroid*	*25% gyroid*
Nozzle temperature	*300°C*	*300°C*	*265°C*
Bed temperature	*80°C*	*80°C*	*100°C*
Chamber temperature	*65°C*	*65°C*	*45°C*
Support interface material	*Nylon*	*Nylon*	*ASA*

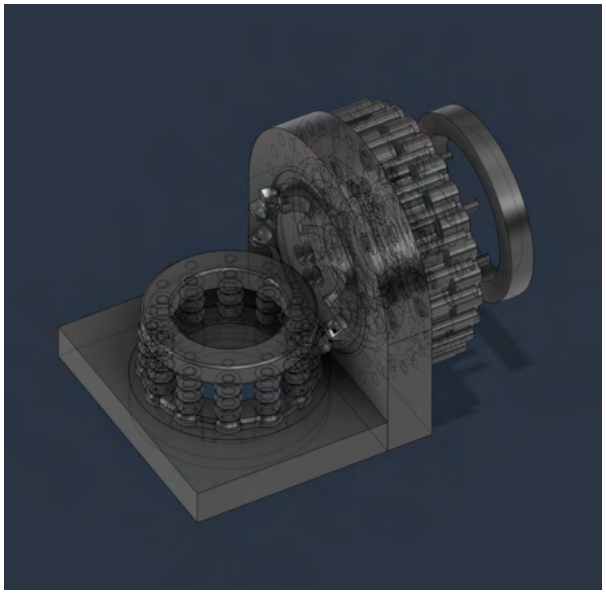
I condition filament using a Sovol SH03 dryer. PET-CF is dried at 80°C for eight hours when necessary, while nylon is dried biweekly at 85°C for twelve hours. I use Magigoo as a release agent for PET-CF and Magigoo PA adhesive for nylon bed adhesion.

Throughout development, I print and check individual components to verify hardware fitment and mating geometry. Hole compensation is applied in the slicer using a 0.09 mm X-Y hole adjustment while retaining nominal CAD dimensions. Where minor surface imperfections interfere with assembly, I locally sand the affected surfaces to achieve the required fit.

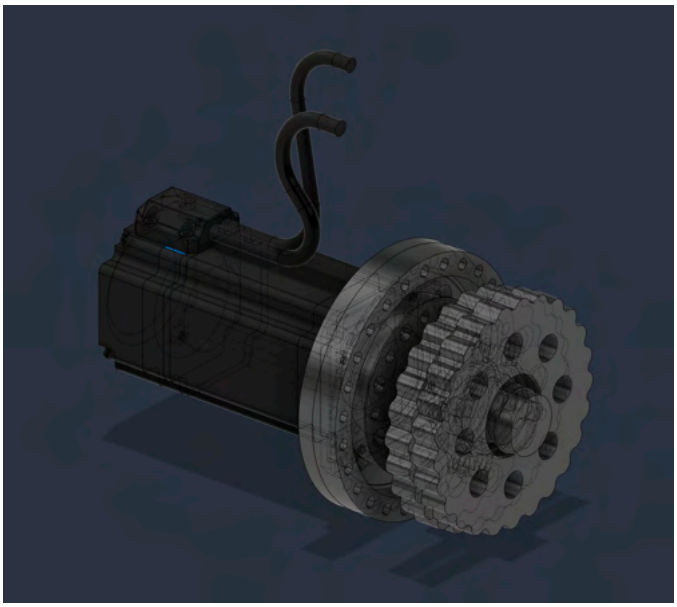
Joint 2 Development and Verification



Three-ring cycloidal prototype: *Initial design intended to conceal the NEMA 24 within the arm for compact, clean packaging. The additional moving components increased mechanical complexity and potential failure points.*



Cycloidal drive with 90° crown-gear input: *Second prototype used a right-angle input to position the motor inside the arm. Hardware size and packaging constraints made the added drivetrain complexity outweigh the benefits.*



Selected direct-input design: *A 25:1 cycloidal reducer connects the motor shaft directly to the eccentric camshaft. This layout prioritizes simpler assembly and fewer potential failure points over concealing the motor within the housing.*

Current Joint 2 Concept

Joint 2 is being developed around a NEMA 24 closed-loop stepper rated at 4.0 N·m and a 25:1 cycloidal reducer. The motor's output shaft connects directly to the eccentric camshaft. An AS5048A magnetic encoder is planned at the gearbox output to measure joint position after reduction.

The design progressed through two earlier prototypes: a three-ring cycloidal drive, followed by a conventional cycloidal drive with a 90° crown-gear input. The crown-gear arrangement introduced excessive mechanical complexity and hardware packaging constraints. I selected the direct-input configuration to simplify the drivetrain and reduce the number of components and interfaces.

Selected cycloidal gears and supporting components have been printed for hardware fitment and dimensional checks. The remaining components are still under development, and the complete gearbox has not yet been assembled or performance tested.

Verification Item	Method	Result
No-load speed	***Measure output revolutions over time using encoder readings or timed rotation.***	***Planned, no measured results***
Backlash	***Restrain the input and measure output angular movement while reversing the applied load.***	***Planned, no measured results***
Output torque	***Apply a known load at a measured lever-arm distance and evaluate holding and driven operation.***	***Planned, no measured results***
Thermal behavior	***Monitor motor, driver, and gearbox temperatures during a defined operating cycle.***	***Planned, no measured results***
Durability	***Run repeated motion cycles under a defined load, then inspect for wear, looseness, and changes in backlash.***	***Planned, no measured results***

Results and Next Steps

Validated performance:

Next design revision: Complete the direct-input gearbox design, print the remaining components, and assemble the prototype. Initial checks will assess hardware fit, component alignment, and rotation without binding before powered testing.

What I learned: Comparing three drivetrain layouts showed me how hardware availability, assembly access, and packaging constraints influenced gearbox design. The direct-input arrangement provides a simpler configuration to manufacture, assemble, and evaluate. I will carry this approach into later joints by checking critical hardware interfaces early and using prototype measurements to guide further revisions.